

Trainings ▾

General Information

The use of quality engine lubricating oils and appropriate oil drain and filter change intervals are critical factors in maintaining engine performance and durability.

Cummins Inc. recommends the use of engine oil that meets the requirements of Cummins® Engineering Standards (CES) 20086, American Petroleum Institute (API) CK-4, **only** in engines with ultra low sulphur diesel (less than 15 ppm). CES 20081 (API CJ-4) may also be used when operating with ultra low sulphur diesel but may require reduced oil drains. When using greater than 15 ppm sulphur fuel, oils meeting CES 20078 (API CI-4), are recommended. CES 20076 or CES 20077 (API CH-4) are also suitable with fuels greater than 15 ppm sulphur. See Fluids for Cummins® Products Service Manual, Bulletin 5411406. Refer to Procedure 378-003 Section 4. (</qs3/pubsys2/xml/en/procedures/00/00-378-003.html>)

The oil suppliers are responsible for the quality and performance of their product.

New Engine Break-in Oils

Special break-in engine lubricating oils are **not** recommended for new or rebuilt Cummins® engines. Use the same type of oil during the break-in as is used in normal operation.

Synthetic or partially synthetic engine oils, however, can **not** be used in a new or rebuilt engine during break-in. Use a standard petroleum-based oil for the first drain interval.

Additional information regarding lubricating oil availability throughout the world is available in the Engine Manufacturers Association Lubricating Oils Data Book for Heavy-Duty Automotive and Industrial Engines. The data book can be ordered from the address below:

Engine Manufacturer's Association

One Illinois Center

2 N Lasalle Street

Chicago, IL U.S.A. 60601

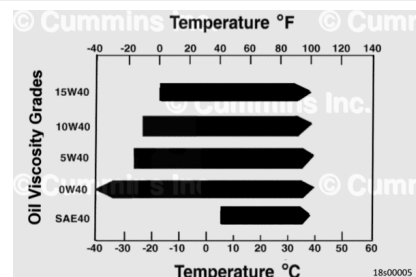
Phone: (312) 644-6610

Cummins Inc. recommends the use of oil based on the most recent API performance categories. For details on lubricating oils for Cummins® engines, contact a Cummins® Authorized Repair Location or see Fluids for Cummins® Products Service Manual, Bulletin 5411406.

Viscosity Recommendations

⚠ CAUTION ⚠

When single-grade oil is used, make sure the oil will be operating within the temperature ranges, as shown.

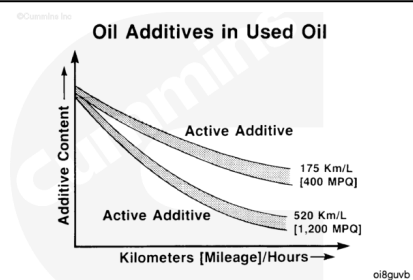


Cummins Inc. recommends the use of 15W-40 multigrade lubricating oils normal conditions. For low ambient temperatures, the diagram on the right shows the preferred oil viscosity grades and temperatures. See Fluids for Cummins® Products Service Manual, Bulletin 5411406, for complete list of recommended Society of Automotive Engineers (SAE) oil viscosity grades.

Single-grade oils can be substituted for short durations until the recommended multigrade is procured. Arctic condition oils are available commercially with better low temperature properties. Consult your supplier.

The primary criterion for selecting an oil viscosity grade is the lowest temperature the oil will experience while in the engine oil sump. Bearing problems can be caused by the lack of lubrication during the cranking and start up of a cold engine when the oil being used is too viscous to flow properly. Change to a lower viscosity grade of oil as the temperature of the oil in the engine oil sump reaches the lower end of the ranges shown in the illustration.

As the engine oil becomes contaminated, essential oil additives are depleted. Lubricating oils protect the engine as long as these additives are functioning properly. Progressive contamination of the oil between oil and filter change intervals is normal. The amount of contamination will vary depending on the operation of the engine, hours or miles on the oil, fuel consumed, and new oil added.



Arctic Operation

CAUTION

The use of a synthetic base oil does not justify extended oil change intervals. Extended oil change intervals can cause engine damage.

If an engine is operated in ambient temperatures consistently below -23°C [-10°F] and there are no provisions to keep the engine warm when it is **not** in operation, use a 0W-40 synthetic engine oil with adequate low temperature properties.

The oil supplier **must** be responsible for meeting the performance service specifications.

Reference Procedure 102-002 (Maintenance Schedule - Oil Drain Intervals) in Section 2 for the use of the Chart Method to allow extended oil change intervals or Service Bulletin, Operation of Diesel Engines in Cold Climates, Bulletin 3379009.

(</qs3/pubsys2/xml/en/bulletin/3379009.html>)